



Daffodil Institute of IT (DIIT)
Suggestion for Statistics and Probability
Course Code: 510217

INTRODUCTION

- # Define statistics. Can a single number of a collection of number be Statistics?
- # Explain some important functions of statistics.
- # Explain the following:
Statistics is the science of average.
- # What is the basic difference between descriptive and inferential statistics?
- # “Statistical methods are most dangerous tools in the hands of the inexpert.”
- # Examine this statement. How are statistics helpful in business and industry? Explain.
- # What are the advantages and disadvantages of statistics?

COLLECTION OF DATA

- # Discuss the different methods of data collection.
- # Define with examples of the followings:
 - (a) Data (b) variable
- # Distinguish between Primary data and secondary data. What precautions would you take before using data from a secondary source?
- # Distinguish between primary and secondary data? Discuss the various methods of collecting primary data. Indicate the situation in which each of these methods should be used.
- # Describe the different methods of collecting data indicating the merits and demerits of each of them.
- # Discuss the various methods of collecting primary data. Indicate the situation in which each of these methods should be used.
- 1. Define level of measurement. What are 4 levels of measurement with examples?

Distinguish between

- (a) Population and sample**
- (b) Discrete variable and continuous variable**
- (c) Discrete data and continuous data**
- (d) quantitative variable and qualitative variable/ attribute**
- (e) qualitative data and quantitative data**
- (f) a schedule and a questionnaire**

What are the methods of collecting primary data

Where you obtain secondary data? What precautions should be taken while using secondary data?

Briefly explain the characteristics of a good questionnaire

What are the precautions of a questionnaire?

PRESENTATION OF DATA

What is classification and tabulation of statistical data?

What is the purpose of classification & tabulation of statistical data?

What is frequency and frequency distribution? Discuss how you will construct a frequency distribution table from raw data.

Explain in brief the different basis of classification of data with suitable examples.

What are the different graphs and charts known to you?

Define with example of the followings:

- (a) class limit (b) class boundary (c) Frequency (d) class intervals (e) mid value (f) Exclusive method (g) Inclusive method**

What is bar diagram? What are the types of bar diagram? Briefly explain.

Distinguish between

- (a) Histogram and bar diagram**
- (b) Histogram and frequency polygon**
- (c) one dimensional and two dimensional diagram**

Problems: 8, 13, 14,15,16,17 [Abdul Aziz (5th Edition, Page-1-67 to 1-79]

MEASURES OF CENTRAL TENDENCY

What do you mean by central tendency? Describe any four measures of central tendency.

- # What are the measures of central tendency? Why are they called measures of central tendency?
- # What are the objectives of averages?
- # What are the properties of a good/ ideal measures of central tendency? Which measure do you think best and why?
- # Give a brief description of different measures of central tendency? Why is arithmetic mean so popular?
- # Define arithmetic mean. Write down the properties of arithmetic mean.
- # Define weighted arithmetic mean. Discuss the importance of weighted arithmetic mean in statistics.
- # Describe the graphical method of determining the median and mode of a frequency distribution.
- # Under what circumstances would it be appropriate to use arithmetic mean, median, mode, geometric mean and harmonic mean.
- # Describe the various methods of measures of central tendency along with their uses, merits and demerits.
- # When mode is ill defined? Write down the modal formula when mode is ill defined.
- # What are the relationship among mean, median and mode?
- # Under what circumstances would it be appropriate to use arithmetic mean, median, mode, geometric mean and harmonic mean. Discuss.
- # Distinguish between median, quartile, deciles and percentiles.
- # All the important proof of central tendency (Which I already given in the Google Classroom.
- Problems: 3, 4,5,6,7,,8,9,10,11,13,16,17,23,24,27,31,32,33,34,35,36,37,38, [Abdul Aziz(5th Edition, Page-2-35 to 2-70]

MEASURES OF DISPERSION/VARIATION

- # What is meant by dispersion? Explain briefly different methods of dispersion.
- # What are the important measures of dispersion? Discuss briefly.
- # What purpose does a measure of variation serve?
- Or, what is the significance of measuring variation?
- # What are the properties of a good measure of dispersion/ variation?
- Or, what are the requisites of a good measure of dispersion/ variation?
- # Distinguish between absolute measure and relative measure.

Define the followings:

**(g) Range (ii) Inter quartile range or quartile deviation (iii) Mean Deviation/
Average Deviation (iv) Standard deviation (v) Coefficient of variation**

Mention the properties of standard deviation.

Explain how measure of central tendency and measure of variation complement each other in describing mass of data.

Under what circumstances range is more meaningful than any other measure of variation?

What is coefficient of variation? What purpose does it serve? Also distinguish between variance and coefficient of variation.

What is standard deviation? Why is the standard deviation usually chosen as a measure of variation?

What are the uses of coefficient of variation in statistical analysis?

Describe the various methods of measuring variation along with their uses, merits and demerits.

All the important proof of dispersion (Which I already given in the Google Classroom.

Problems: 12, 13, 15, 17, 21, 22, 23, 29, 31, 34, 35 [Abdul Aziz (5th Edition, Page-3(1)-42 to 3(1) 73]

Skewness, Moments and Kurtosis

1. What is moment? Define central moments and raw moments?
2. Distinguish between central moments (moments about mean) and raw moments (moments about arbitrary value).
3. What do you mean by moments, skewness and kurtosis? Distinguish between dispersion and skewness and point out the methods of measuring skewness.
4. Define moments. How can you find out skewness and kurtosis of a distribution from moments about the mean?
5. Explain clearly how the moments help in describing the characteristics of a frequency distribution.
6. Distinguish between variation and skewness and point out the various methods of measuring skewness.

7. Distinguish between skewness and kurtosis. How are the measures of skewness and kurtosis used in business decision making? Discuss.
8. Explain different kinds of skewness and kurtosis with the help of graphs.
9. Distinguish between skewness and kurtosis.
10. Explain the term 'skewness'. What purpose does a measure of skewness serve?
Comment on some of the well known measures of skewness.
11. **# important proof of moments, skewness and kurtosis (Which I already given in the Google Classroom).**

Problems: 1, 2, 3, 4, 6, 8, 9, 10, 11[Abdul Aziz (5th Edition, Page-3(2)-11 to 3(2) 20]

Correlation and regression analysis

1. What do you mean by bivariate distribution? Discuss the concept of correlation and regression and point out their usefulness in dealing with business problem.
2. Distinguish between correlation and regression.
3. Interpret the followings:
(i) $r = -1$ (ii) $r = 0$ (iii) $r = 1$ (iv) $r = 0.5$ (v) $r = 0.3$ (vi) $r = 0.9$ (viii) $r = -0.25$
(ix) $r = -0.85$ (x) $r = 0.03$
4. What is coefficient of determination? $r^2 = 0.36$. What does it indicates?
5. What is a scatter diagram? Discuss different nature of correlation with the help of Scatter diagram.
6. Distinguish the followings with example: (i) Positive and negative correlation
(ii) Dependent and independent variables (iii) Karl Pearson's correlation coefficient and Spearman's rank correlation coefficient.
7. Describe the methods of least squares and show how it can be used to fit a linear regression.
8. What is Spearman's rank correlation coefficient? Bring out its usefulness. How does the coefficient differ from Karl Pearson's coefficient of correlation?
9. Define Spearman's rank correlation coefficient. What are the merits and limitations of rank correlation coefficient?
10. What is regression analysis? Discuss the importance / uses of regression analysis in business.

11. What is meant by regression coefficient? Mention the properties of regression coefficients. Interpret the followings:

$$(i) b_{yx} = 0.8 \quad (ii) b_{xy} = 0.6$$

12. Define the coefficient of correlation. Write down its properties.

13. Explain the meaning and significance of the term correlation. Discuss the different types of correlation.

All the important proof of Correlation and regression(Which I already given in the Google Classroom.)

14. Define the followings:

(i) Probable error (ii) Standard error (iii) Standard error of correlation coefficient

Problems: Correlation Analysis: Already given in the Google Classroom

Regression Analysis: Already given in the Google Classroom

N.B: **Solve at least last 4 years NU questions

****Try to solve all the problems of Google Classroom and then try to solve problems of all the chapter which I given in the above.**

Good Luck